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The Grassroots-to-Mastery Roadmap

Phase 1: The Harmonic Dilemma

Core Saturation, Triplen Overflow, and Winding Selections

Module Objective:

To transition from linear ideal transformers to real-world non-linear magnetic cores. You will master the Fourier mathematics behind magnetizing current distortion, mathematically prove the zero-sequence nature of triplen harmonics, and geometrically deduce why ungrounded Star-Star ($Y - Y$) connections lead to catastrophic insulation failure.

Pedagogical Sequence Framework

Phase 1 of 6

May 22, 2026

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1 The Non-Linear Magnetic Core and Excitation

Cognitive Thought Process & Learning Sequence

1. In basic circuit theory, an inductor (L) draws a perfect sine wave of current to make a perfect sine wave of flux. This assumes L is constant.
2. Real transformers use silicon steel. Steel has a limit to how much flux it can hold. This is the B-H curve (Saturation).
3. We must explore what happens when we demand a perfect sine wave of flux from a core that is nearing its saturation limit.

1.1 The Requirement for Sinusoidal Flux

The ultimate goal of a power transformer is to deliver a perfectly clean, undistorted sinusoidal voltage to the consumer. According to Faraday's Law of Electromagnetic Induction, the induced phase voltage (e) is the negative time derivative of the magnetic flux (ϕ) multiplied by the number of turns (N):

$$e(t) = -N \frac{d\phi}{dt} \quad (1)$$

If the induced voltage $e(t)$ must be a pure cosine/sine wave, then mathematically, the flux $\phi(t)$ **must** also be a pure sine wave. There is no negotiation on this physical law.

1.2 The Magnetizing Current Distortion

To generate this perfectly sinusoidal flux $\phi(t) = \Phi_{\max} \sin(\omega t)$, the primary winding must draw a magnetizing current (i_m) from the supply.

In an ideal core (air core), flux is directly proportional to current ($\Phi \propto i$). However, in a ferromagnetic steel core, the relationship is dictated by the non-linear **B-H hysteresis curve**.

As the sinusoidal flux approaches its peak value during each half-cycle, the core begins to saturate. In the saturation region, the core becomes "stiff." To push the flux up by just a tiny amount to reach its required sinusoidal peak, the winding must draw a **disproportionately massive amount of current**.

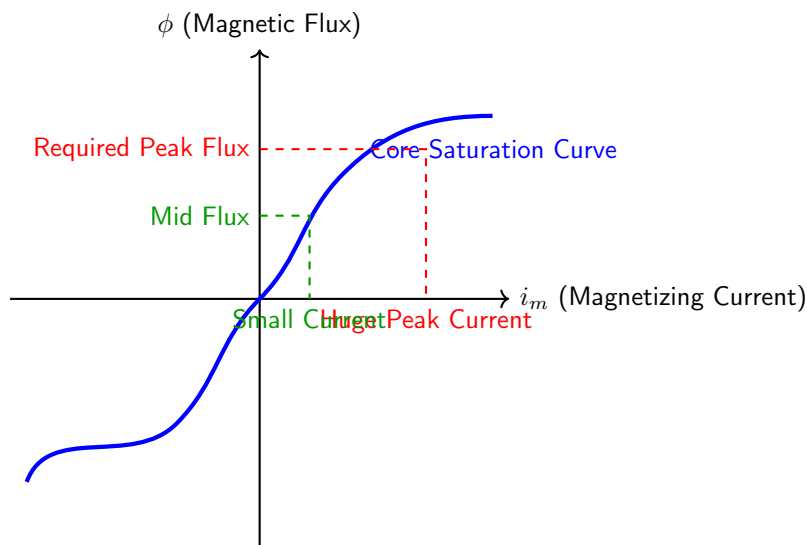


Figure 1: The B-H Curve Mapping. Notice how a linear increase in flux near the top requires an exponential increase in current.

Because of this saturation, the magnetizing current cannot be a smooth sine wave. It must spike sharply at the peaks to force the flux to remain sinusoidal. This results in a highly **peaked, non-sinusoidal magnetizing current waveform**.

2 Fourier Analysis and Triplen Harmonics

Cognitive Thought Process & Learning Sequence

1. We have a weird, pointy, peaked current wave. How do we mathematically analyze it?
2. We use Fourier Series. Any repeating distorted wave is just a stack of pure sine waves at different frequencies (harmonics).
3. Because the peaked wave is symmetrical (top half looks like bottom half), it only contains ODD harmonics (3rd, 5th, 7th...).
4. The 3rd harmonic is the biggest and most problematic. We call multiples of 3 "Triplen" harmonics.

If we apply a Fourier series expansion to our highly peaked magnetizing current $i_m(t)$, we get:

$$i_m(t) = I_1 \sin(\omega t) - I_3 \sin(3\omega t) + I_5 \sin(5\omega t) - I_7 \sin(7\omega t) + \dots \quad (2)$$

- $I_1 \sin(\omega t)$ is the **Fundamental component** (50 Hz).
- $I_3 \sin(3\omega t)$ is the **Third Harmonic component** (150 Hz). It is the largest harmonic and is responsible for creating the sharp "peak" in the current waveform.

2.1 The Zero-Sequence Nature of Triplen Harmonics

This is the most critical mathematical proof in this phase. In a balanced three-phase system, the fundamental currents are displaced by 120° in time:

$$i_{a1}(t) = I_1 \sin(\omega t)$$

$$i_{b1}(t) = I_1 \sin(\omega t - 120^\circ)$$

$$i_{c1}(t) = I_1 \sin(\omega t + 120^\circ)$$

Now, let us calculate the phase angles for the **Third Harmonic** components. The frequency is 3ω , which means we must multiply the fundamental phase shifts by 3:

$$i_{a3}(t) = I_3 \sin(3\omega t) \tag{3}$$

$$i_{b3}(t) = I_3 \sin(3(\omega t - 120^\circ)) = I_3 \sin(3\omega t - 360^\circ) \equiv I_3 \sin(3\omega t) \tag{4}$$

$$i_{c3}(t) = I_3 \sin(3(\omega t + 120^\circ)) = I_3 \sin(3\omega t + 360^\circ) \equiv I_3 \sin(3\omega t) \tag{5}$$

Conclusion: Because subtracting or adding 360° results in the exact same phase angle, the third harmonic currents in Phase A, Phase B, and Phase C are **perfectly in phase with each other**. They hit their peaks at the exact same millisecond. In Symmetrical Components theory, phasors that are identical in magnitude and phase are called **Zero-Sequence Components**. This applies to all Triplen harmonics (3^{rd} , 9^{th} , 15^{th} , etc.).

3 The Ungrounded Star-Star ($Y - Y$) Disaster

MANDATORY MULTIMEDIA INTEGRATION: YOUTUBE LECTURE

Video Topic: Transformer Harmonics and Neutral Shift

Search Query / Link: https://www.youtube.com/results?search_query=transformer+harmonics+nptel

Required Focus: Search for visual proofs of why ungrounded systems fail under third harmonic conditions. Watch how the neutral point oscillates.

Cognitive Thought Process & Learning Sequence

1. We just proved that 3^{rd} harmonic currents are in-phase ($i_{3a} = i_{3b} = i_{3c}$).
2. We also know Kirchhoff's Current Law (KCL): The sum of currents entering a star neutral point must be zero.
3. If the neutral point is ungrounded (no neutral wire back to the source), KCL forces $3 \times i_3 = 0$.
4. Therefore, the 3^{rd} harmonic current cannot physically flow.
5. What happens to the core flux if we deny it the peaked current it demands? It goes flat!

3.1 The Physics of the Flat-Topped Flux

In an ungrounded $Y - Y$ connection, the total line current must sum to zero at the star point:

$$i_A + i_B + i_C = 0$$

Since $i_{3a} = i_{3b} = i_{3c} = i_3$, substituting this into KCL gives:

$$3i_3 = 0 \implies i_3 = 0$$

Because the circuit physically blocks the 3rd harmonic current from flowing, the magnetizing current remains a pure fundamental sine wave (50 Hz only).

However, we established in Section 1 that a pure sine wave of current is **not enough** to drive the core to a sinusoidal flux peak due to saturation. Because the current lacks the sharp peaks required to push through the saturated "stiff" region, the resulting magnetic flux fails to reach its peak. The top of the flux wave gets "chopped off." This creates a **Flat-Topped Flux Waveform**.

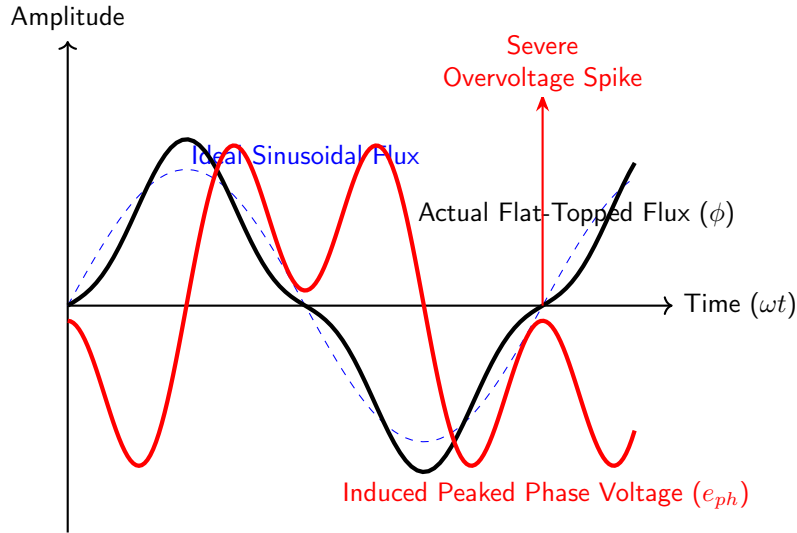


Figure 2: The Consequence of Missing Triplen Currents: The flat-topped flux induces a highly peaked, dangerous phase voltage.

3.2 Mathematical Proof of Voltage Breakdown

A flat-topped flux wave can be mathematically represented as a fundamental wave minus a 3rd harmonic flux component:

$$\phi(t) = \Phi_1 \sin(\omega t) - \Phi_3 \sin(3\omega t) \quad (6)$$

To find the induced phase voltage across the transformer windings, we apply Faraday's Law ($e = -N \frac{d\phi}{dt}$):

$$e(t) = -N \frac{d}{dt} [\Phi_1 \sin(\omega t) - \Phi_3 \sin(3\omega t)] \quad (7)$$

$$e(t) = -N\omega\Phi_1 \cos(\omega t) + N(3\omega)\Phi_3 \cos(3\omega t) \quad (8)$$

$$e(t) = E_{1\max} \cos(\omega t) + \mathbf{3}E_{3\max} \cos(3\omega t) \quad (9)$$

The Danger: Notice the multiplier coefficient **3** that drops down from the chain rule during differentiation. The 3rd harmonic voltage is amplified by a factor of 3! When you differentiate

a flat-topped wave, the slope changes violently. This results in a phase voltage waveform that is extremely peaked (often 1.5 to 1.8 times higher than the nominal RMS peak).

These massive voltage spikes severely stress the inter-turn dielectric insulation of the phase windings, leading to premature aging, corona discharge, and eventual catastrophic short-circuit failure.

4 The Delta (Δ) Winding Savior

Cognitive Thought Process & Learning Sequence

1. We know ungrounded $Y - Y$ is a disaster because 3rd harmonic currents can't flow.
2. We need a way to let these co-phasor currents flow, without sending them out onto the transmission lines (which would cause telecom interference).
3. What if we connect the windings in a closed loop? Enter the Delta (Δ) connection.

If either the primary or the secondary winding is connected in a Delta (Δ) configuration, the problem is instantly solved.

In a Delta connection, the end of one phase winding is connected to the start of the next, forming a closed triangular mesh. Because the 3rd harmonic EMFs induced in the three phases are exactly in-phase (zero-sequence), they do not cancel each other out around the closed loop. Instead, they add together sequentially:

$$e_{\text{loop},3} = e_{3a} + e_{3b} + e_{3c} = 3e_3$$

This net 3rd harmonic voltage drives a continuous circulating 3rd harmonic current entirely within the closed Delta loop:

$$I_{\Delta,3\text{rd}} = \frac{3e_3}{3Z_{\text{phase},3\text{rd}}} = \frac{e_3}{Z_{\text{phase},3\text{rd}}} \quad (10)$$

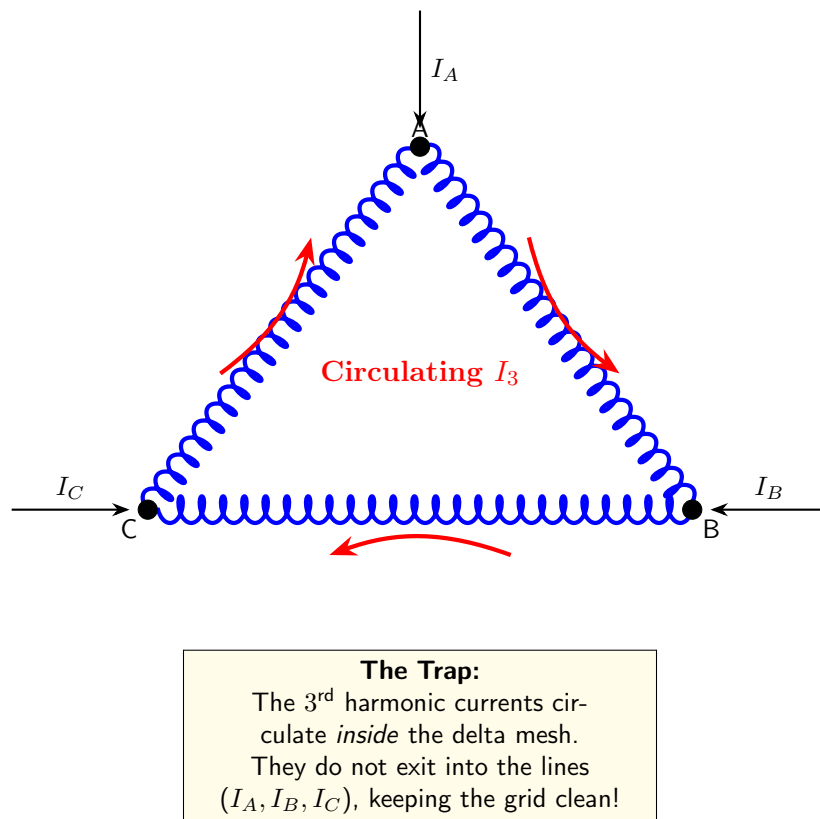


Figure 3: The Delta Connection traps and circulates the required 3rd harmonic magnetizing currents locally.

Because the core is now successfully supplied with the required 3rd harmonic magnetizing current (drawn from this circulating loop rather than the external lines), the core flux is restored to a pure, smooth sine wave. The phase voltages are saved from distortion.

This is exactly why **Delta-Star** ($\Delta - Y$) or **Star-Delta** ($Y - \Delta$) configurations are universally preferred in power distribution networks over ungrounded Star-Star.

5 The 3-Limb Core: Tank Heating Phenomenon

Cognitive Thought Process & Learning Sequence

1. We know triplen fluxes (Φ_3) are zero-sequence (all in phase).
2. Imagine a 3-limb core. At time t , Phase A flux is pointing UP. Phase B flux is pointing UP. Phase C flux is pointing UP.
3. The fluxes hit the top yoke. Where do they go? There is no 4th iron leg to travel down!
4. They have to jump out of the core, pass through the oil, and return down the steel walls of the transformer tank.
5. Changing magnetic flux + solid steel plate = Massive Eddy Currents (Heating).

If an ungrounded $Y - Y$ transformer is built using three independent single-phase units (a transformer bank) or a 5-limb shell-type core, the flat-topped flux problem generates 3rd harmonic flux components (Φ_3). In those core types, the Φ_3 flux can easily return through the separate, independent iron legs.

However, the most common and economical transformer design is the **3-Phase, 3-Limb Core-Type**. This specific geometry creates a unique and severe problem.

In a 3-limb core, the three vertical legs share common top and bottom yokes. The triplen harmonic fluxes ($\Phi_{3a}, \Phi_{3b}, \Phi_{3c}$) are co-phasor. Therefore, at any instant, they are all attempting to flow in the same direction (e.g., vertically upwards) through all three limbs.

Because they are all flowing upward, they sum together at the top yoke:

$$\Phi_{3\text{total}} = \Phi_{3a} + \Phi_{3b} + \Phi_{3c} = 3\Phi_3$$

Since there is no 4th return iron leg, this massive sum of triplen flux cannot return through the core. It is forced to leave the high-permeability iron, traverse the high-reluctance transformer oil, and return through the magnetic **mild steel walls of the transformer tank**.

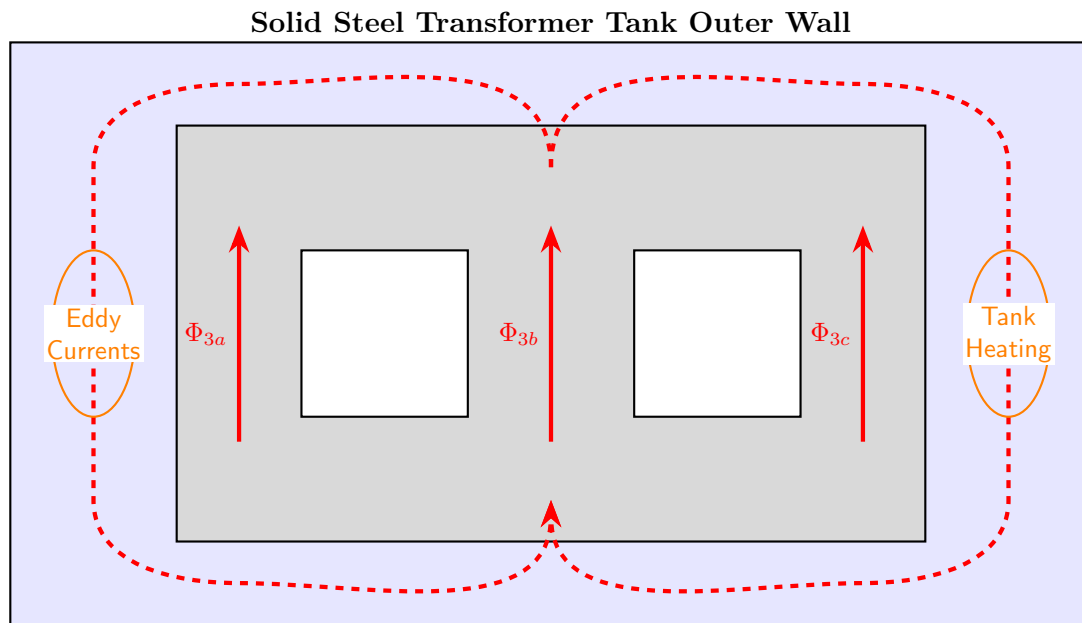


Figure 4: Triplen Flux Leakage in a 3-Limb Core. The fluxes escape the yokes and induce massive eddy currents in the tank walls.

The transformer tank is made of solid, thick, unlaminated steel. When high-frequency (150 Hz) alternating flux cuts through a solid conductor, it acts as a generator. According to Faraday's Law, it induces localized voltages within the steel plate. Because the steel is a continuous sheet, these voltages drive massive, swirling **Eddy Currents**.

These eddy currents result in massive I^2R power losses entirely within the tank walls. This phenomenon, known as **Stray Tank Heating**, severely degrades the transformer's efficiency, overheats the cooling oil, and can lead to localized melting or fire hazards. This structural flaw is why a Delta tertiary winding is absolutely mandatory if a 3-limb core is used in a Star-Star configuration.

6 Phase 1 Review and Self-Evaluation

Cognitive Thought Process & Learning Sequence

You have now mastered the most conceptually difficult part of three-phase transformer theory. Verify your cognitive map before advancing to Phase 2 (Spatial Vector Topology).

6.1 Theoretical Competency Checklist

1. **The Saturation Problem:** Can you explain why a purely sinusoidal flux demands a highly peaked, distorted magnetizing current?

2. **Zero-Sequence Mathematics:** If an examiner asks you to prove that 3rd harmonics are in-phase, can you write out the $3 \times 120^\circ = 360^\circ$ mathematical proof?
3. **The KCL Blockade:** Why does an ungrounded neutral point physically prevent 3rd harmonic currents from flowing?
4. **The Derivative Threat:** Mathematically, why does a flat-topped flux wave create a severely peaked, spiky phase voltage wave? (Hint: The chain rule multiplier 3ω).
5. **The Delta Solution:** Draw the delta loop. Explain how it traps the zero-sequence currents inside the transformer.
6. **Tank Heating Mechanism:** Contrast a 5-limb core with a 3-limb core. Why does only the 3-limb core cause eddy current heating in the solid steel outer tank?

You have successfully completed Phase 1. You possess the theoretical foundations of harmonic distortion. You are now prepared to tackle Phase 2: Spatial Vector Topology and Clock Conventions.